The Renovation Roadmap

Finding Tomorrow's Gentrification in Today's Building Permits

Spatiotemporal Pattern Discovery of Urban Investment Hotspots in Saint John, New Brunswick

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Code and Data: https://github.com/MalekKaroui/renovation-roadmap

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Abstract

Urban gentrification usually becomes obvious only after property values have already increased and neighborhood change is well underway. This makes cities reactive rather than proactive. In this project, I analyze building permit data from Saint John, New Brunswick (2015–2025) to see whether investment patterns can provide an earlier warning signal. I use three main analytical approaches: K-Means clustering, DBSCAN, and Getis-Ord spatial hotspot analysis. I also develop a new metric called the Investment-Assessment Ratio (IAR), which compares recent permit investment to current assessed property value in the same area. When this ratio is high in a low-value area, it suggests that investment is moving in faster than official property values have adjusted. Using this approach, I identified 15 grid cells that appear to show pre-gentrification signals, one statistically significant hotspot, and one grid cell with 71% annual growth over a 10-year period. The project suggests that building permits can work as a practical leading indicator of neighborhood change and could help municipalities respond earlier to infrastructure and housing pressures.

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1 Introduction

1.1 The Problem

Cities often respond to neighborhood change too late. By the time gentrification becomes visible in tax assessments, property sale prices, or census data, it is already well underway. Residents may already be facing rising rents, property tax pressure, and displacement, while the city may be dealing with unexpected infrastructure demand.

This project starts from a simple idea: building permits may reveal neighborhood change earlier than other indicators. If a low-value area begins receiving high-value permits, that may be a sign that capital is moving in before official property values catch up. If this pattern can be detected systematically, it may provide cities with a useful early warning system.

1.2 Research Question

This project asks the following question:

> *Can building permit data be used to identify early signs of gentrification in Saint John before those changes appear in property assessments?*

To answer this, I focus on four smaller questions:

- Where are high-value permits concentrated in Saint John?
- Are there areas where investment appears to be arriving faster than assessment values?
- How do different spatial methods describe the same city?
- Which areas seem to be changing most quickly over time?

1.3 Motivation

The motivation for this project comes from the gap between when neighborhood change starts and when cities officially recognize it. Traditional indicators such as tax assessments, sale prices, and census data are useful, but they usually reflect change after it has already happened. Building permits are different. They are filed before construction is finished, before sales happen, and before property tax values are updated. That makes them a promising early warning signal.

The main contribution of this project is a practical framework for identifying early-stage gentrification in Saint John using building permits, spatial analysis, and a new Investment-Assessment Ratio metric.

1.4 Why This Project Is Interesting

I found this project interesting for two reasons. First, it uses local New Brunswick data instead of a generic dataset. Second, it tries to treat permits as a *leading indicator* of change instead of just a record of construction activity. That makes the project more useful for real planning decisions.

Saint John is also a good case study because it is not a very large city. Most gentrification research focuses on large metropolitan areas like Toronto, Vancouver, New York, or Chicago. Saint John offers a smaller urban setting where neighborhood change may happen in a more scattered way.

2 Background and Related Work

2.1 Gentrification as Capital Movement

A classic starting point for discussing gentrification is Smith's work on the "rent gap" [1]. His argument is that gentrification is not simply about people moving back into cities, but about capital returning to undervalued urban land. Developers invest when they believe the gap between current value and potential value is large enough to make redevelopment profitable.

This idea is directly relevant to my project. If permits represent investment decisions, then they may reveal where developers believe that value gap exists.

2.2 How Gentrification Is Usually Measured

A common problem in gentrification research is timing. Researchers often rely on:

- census data,
- property sale prices,
- assessed property values,
- or demographic change over long time periods.

These are useful, but they tend to show change after it is already happening. Ding et al. [2] showed that building permits can predict later increases in property values. That was important for my project because it supports the idea that permits can be used as an earlier signal.

2.3 Spatial Hotspot Analysis

Getis and Ord [3] introduced a spatial statistic that helps identify whether high or low values are clustered in a way that is unlikely to be random. The statistic is:

$$G_i^* = \frac{\sum_j w_{ij} x_j - \bar{X} \sum_j w_{ij}}{S \sqrt{\frac{n \sum_j w_{ij}^2 - (\sum_j w_{ij})^2}{n-1}}} \tag{1}$$

where:

- x_j is the value at location j,
- w_{ij} is the spatial relationship between locations i and j,
- $\bar{X}$ is the global mean,
- S is the standard deviation,
- and n is the number of locations.

In simple terms, the statistic asks whether one location is surrounded by unusually high or unusually low values. This is useful because it moves the analysis beyond visual inspection and gives a formal statistical result.

2.4 Why I Used a Grid

My original plan was to use neighborhood boundaries. However, I was not able to get a clean and reliable neighborhood boundary file from the Saint John open data sources. Rather than forcing a weak boundary layer into the project, I switched to a 1 km by 1 km regular grid.

This decision was also supported by prior work. Behnisch and Ultsch [4] show that grid-based spatial analysis can be effective for urban data mining because it avoids irregular polygon sizes and outdated administrative boundaries.

3 Data Collection and Preparation

3.1 Datasets Used

3.1.1 Building Permits

The main dataset came from the City of Saint John Open Data Catalogue [5]. It contained 30,287 permit records. Important attributes included:

- `BPEVAL`: permit valuation,
- `BPATYP`: permit type,
- `BPPISD`: issue date,
- `X_Long_East` and `Y_Lat_North`: coordinates,
- `ENCIVSTNAM`: street address.

3.1.2 Property Assessments

The second major dataset was a GeoJSON file containing 76,394 property assessments, with:

- `CURR_ASSMT`: current assessed value,
- geometry information for parcel boundaries,
- and property description fields.

These records provided the baseline against which permit investment could be compared.

3.2 Cleaning the Permit Data

The raw permit data needed substantial cleaning before it could be used.

3.2.1 Dates

The `BPPISD` field stored dates as 6-digit integers. For example, `251030` represented October 30, 2025. I wrote a custom parser to convert these values to dates. In the process, I found 5,777 invalid records and removed them.

3.2.2 Time Filter

I restricted the analysis to permits from 2015 to 2025. This reduced the working dataset from 24,510 to 23,514 records and kept the analysis focused on recent urban change.

3.2.3 Coordinates

At first, I assumed that `POINT_X` and `POINT_Y` were the main coordinate fields, but these turned out not to be the correct longitude and latitude values for this analysis. The actual WGS84 coordinates were stored in `X_Long_East` and `Y_Lat_North`. After switching to those columns, I validated coordinates against a Saint John bounding box and removed 45 invalid records.

3.2.4 Outliers and Duplicates

Valuation cleaning removed 7,500 problematic records, including zero-dollar permits and extreme outliers. Then I found 14,871 duplicate permit records using address, date, and value combinations. After removing these, the final cleaned permit dataset contained 1,098 records.

3.3 Final Permit Dataset

The cleaned permit data had the following characteristics:

Table 1: Summary Statistics for Cleaned Building Permit Data

Metric	Value
Raw permits	30,287
Final cleaned permits	1,098
Total investment represented	$255,986,293
Mean permit value	$233,139
Median permit value	$28,000
Grid cells with permit activity	151

The large difference between the mean and median indicates a strongly right-skewed distribution.

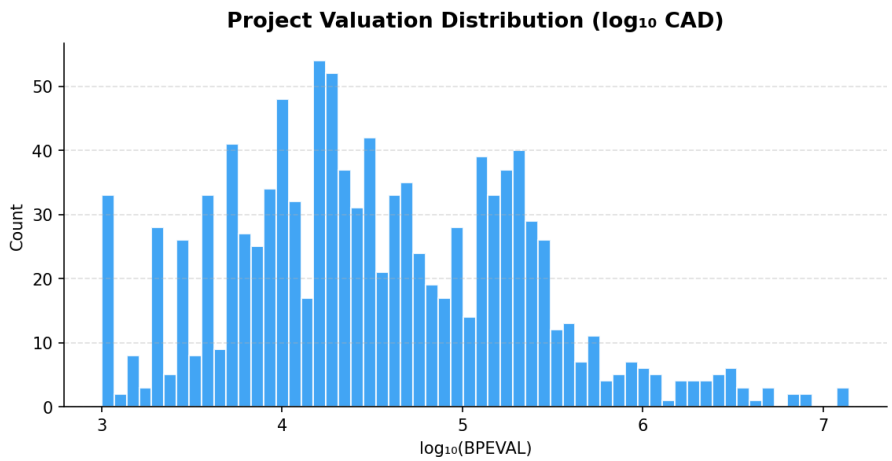

Figure 1: Distribution of permit valuations on a log scale. Most permits are relatively small, while a smaller number of very large permits create a long right tail.

3.4 Spatial Framework

All coordinates were reprojected to UTM Zone 20N for distance-based analysis. I then created a regular 1 km by 1 km grid over the city.

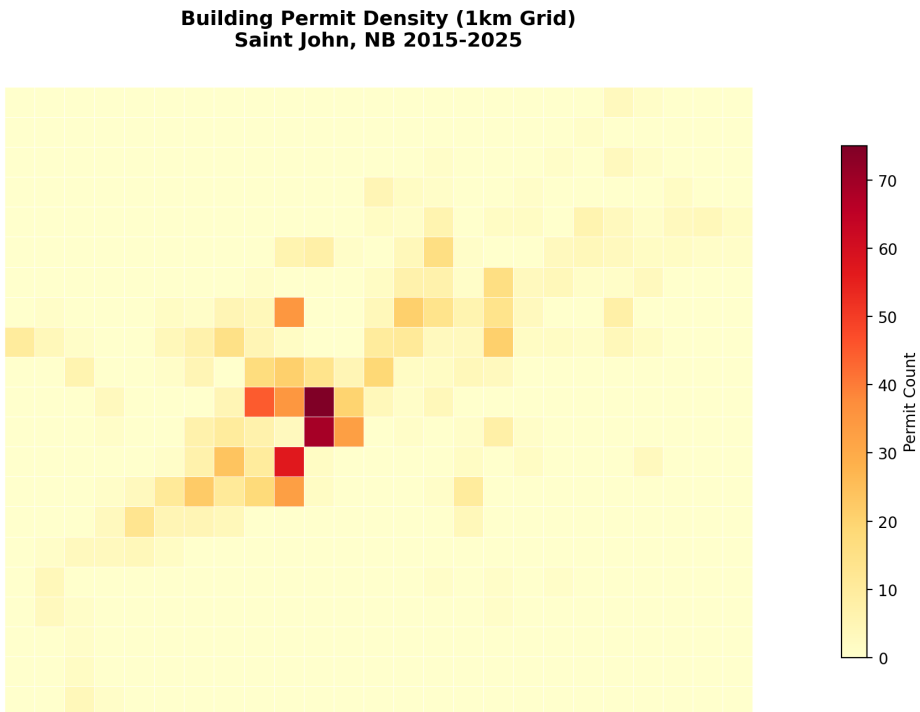

Figure 2: Permit density by 1 km grid cell. Permit activity is clearly concentrated in some parts of the city and sparse in others.

This produced:

- 525 total grid cells,
- 151 active grid cells with permit activity,
- and a consistent spatial unit for later analysis.

4 Methods

4.1 The Main Idea: Investment-Assessment Ratio

The most important idea in the project is that building permits may capture neighborhood change before property assessments do. This led me to define the Investment-Assessment Ratio (IAR):

$$\text{IAR} = \frac{\text{mean permit value}}{\text{mean assessed property value}} \tag{2}$$

If this ratio is high, and the baseline assessment is still low, it suggests that investment is entering the area faster than official values have adjusted.

4.2 The 8-Stage Analysis Pipeline

The final implemented pipeline has 8 stages:

1. **Data preprocessing**
2. **Spatial analysis and grid creation**
3. **K-Means clustering**
4. **DBSCAN clustering**
5. **Getis-Ord hotspot analysis**
6. **Temporal analysis**
7. **Assessment integration and IAR calculation**
8. **Final synthesis and visualization**

In addition to the main pipeline, I also created an interactive Streamlit dashboard for presentation and exploration of results. I also created a Windows batch runner to automate the full workflow and improve the readability of terminal output during live demonstrations.

4.3 K-Means Clustering

For K-Means, I used four features:

- projected X coordinate,
- projected Y coordinate,
- $\log_{10}$ of permit value,
- permit year.

I tested values of k from 2 to 15 using the elbow method and silhouette scores. Both suggested $k = 6$ as a reasonable choice.

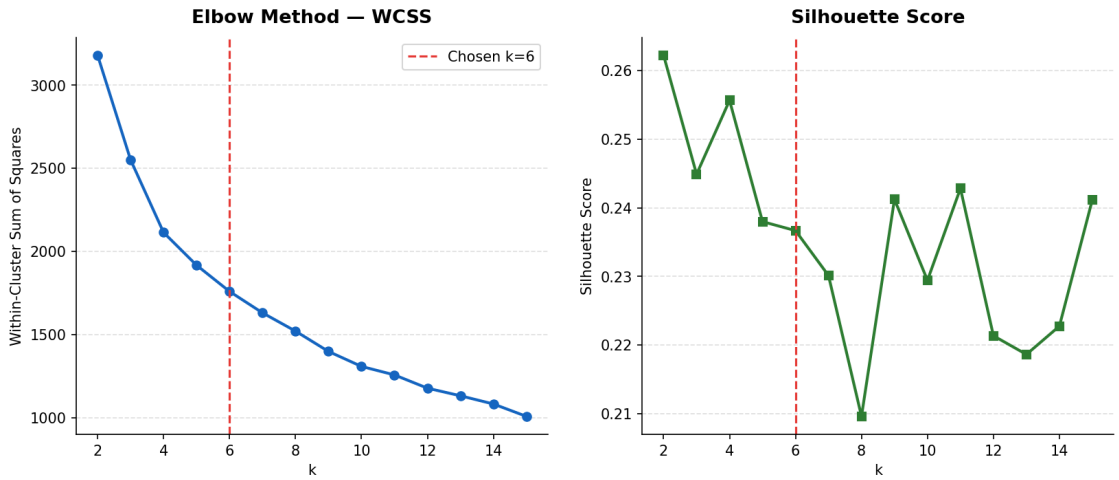

Figure 3: K-Means parameter selection. The elbow curve and silhouette scores both support using 6 clusters.

The most important K-Means result was Cluster 4:

- 244 permits,
- mean value $841,793.53,
- median value $211,000.

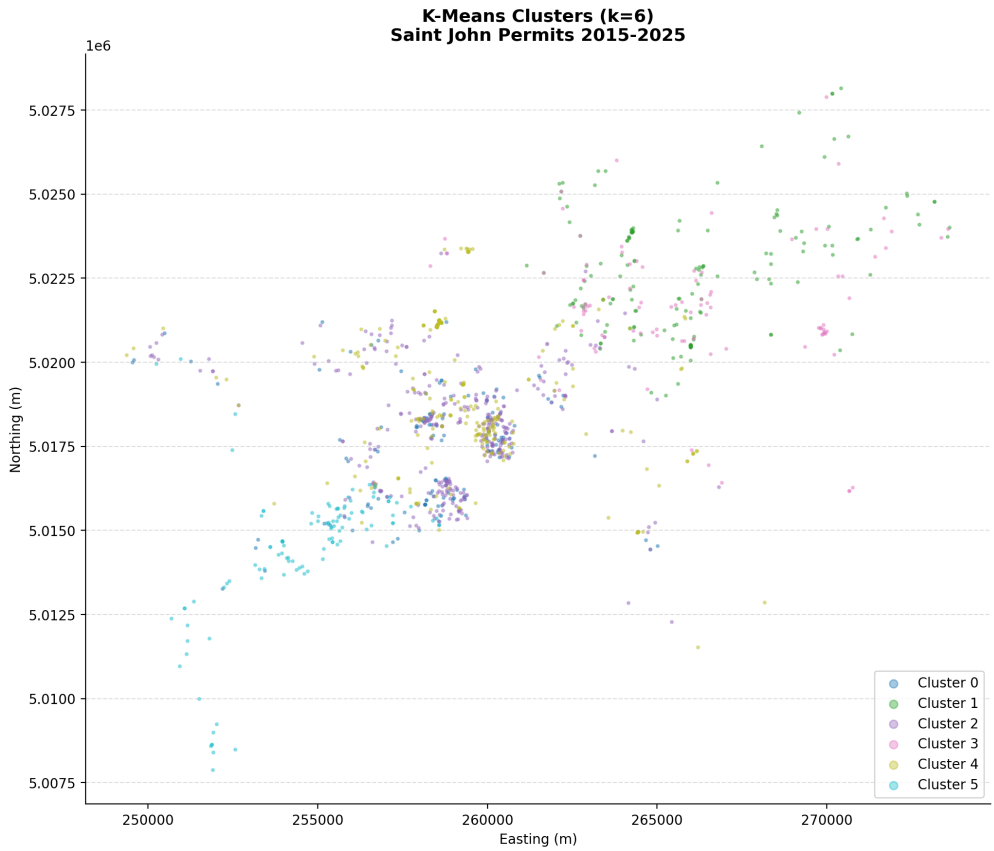

Figure 4: Spatial distribution of K-Means clusters. The highest-value cluster is concentrated in the urban core.

4.4 DBSCAN

I used DBSCAN as a second clustering method because it does not require choosing a fixed number of clusters in advance and can label outliers as noise. The estimated epsilon value was 2.1565.

The final DBSCAN result was:

- 1 main cluster,
- 1 noise point,
- 0.1% of points labeled as noise.

This result was not visually exciting, but it was useful. It suggested that permit activity in Saint John is not split into several strongly separated spatial groups.

4.5 Getis-Ord Hotspot Analysis

I applied the Getis-Ord G_i^* statistic at the grid-cell level using Queen contiguity weights.

The main result was:

- 1 statistically significant hotspot,
- GRID_04_16,
- $G_i^* = 2.009144$,
- $p = 0.026$.

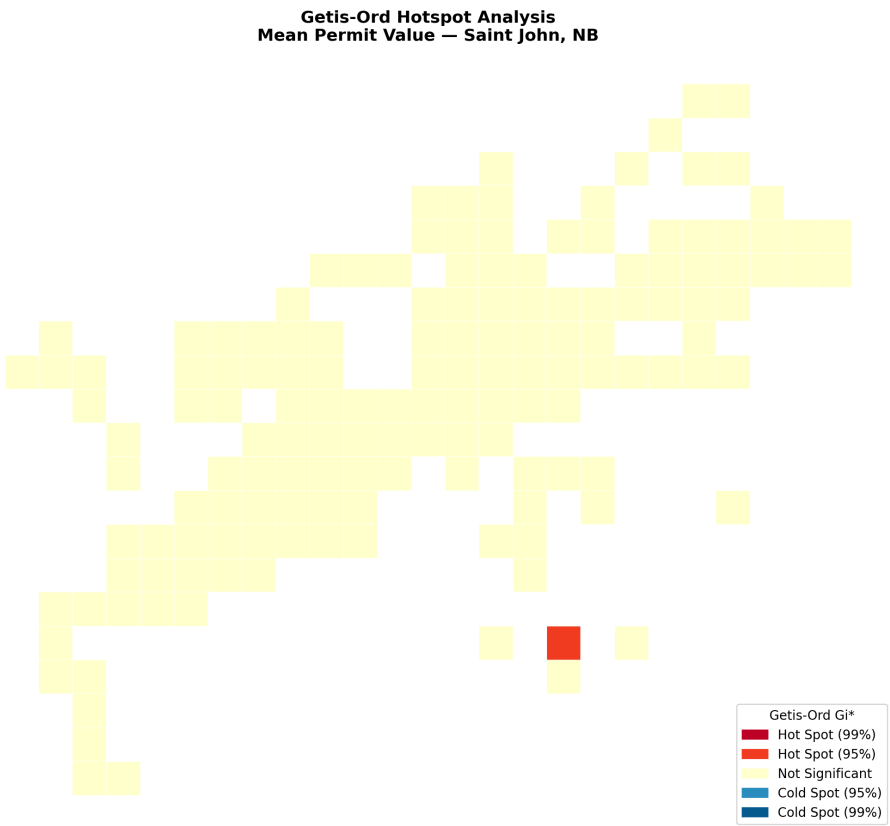

Figure 5: Getis-Ord hotspot analysis. One significant hotspot was found, while most cells were not statistically significant.

4.6 Temporal Analysis

I divided the data into multiple time windows and calculated growth rates for each active cell. The main temporal metric was compound annual growth rate (CAGR):

$$\text{CAGR} = \left(\frac{\text{value}_{\text{last}}}{\text{value}_{\text{first}}}\right)^{1/10} - 1 \tag{3}$$

The main finding here was:

- 1 emerging grid cell,
- GRID_10_10,

- CAGR = 0.710492, or about 71%.

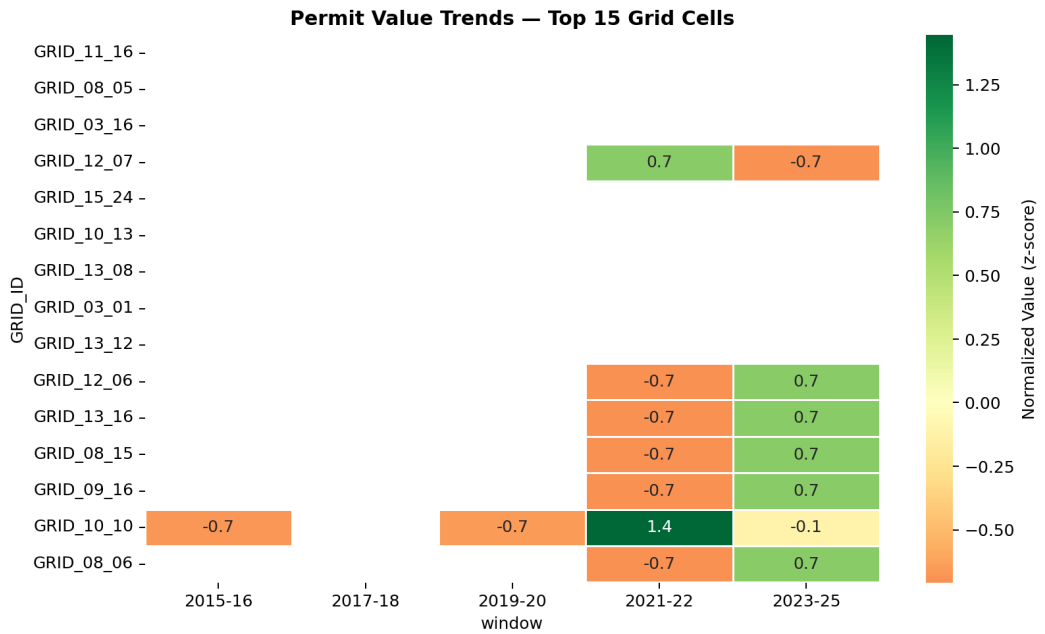

Figure 6: Temporal heatmap for the top active cells. GRID_10_10 shows the strongest upward pattern over time.

4.7 Assessment Integration and IAR

I spatially joined property assessments to the same 1 km grid and calculated mean assessed value for each relevant cell. This allowed me to compute IAR.

The classification rule for a pre-gentrification signal was:

- IAR in the top quartile,
- mean assessed value in the bottom quartile.

This identified 15 pre-gentrification signal cells.

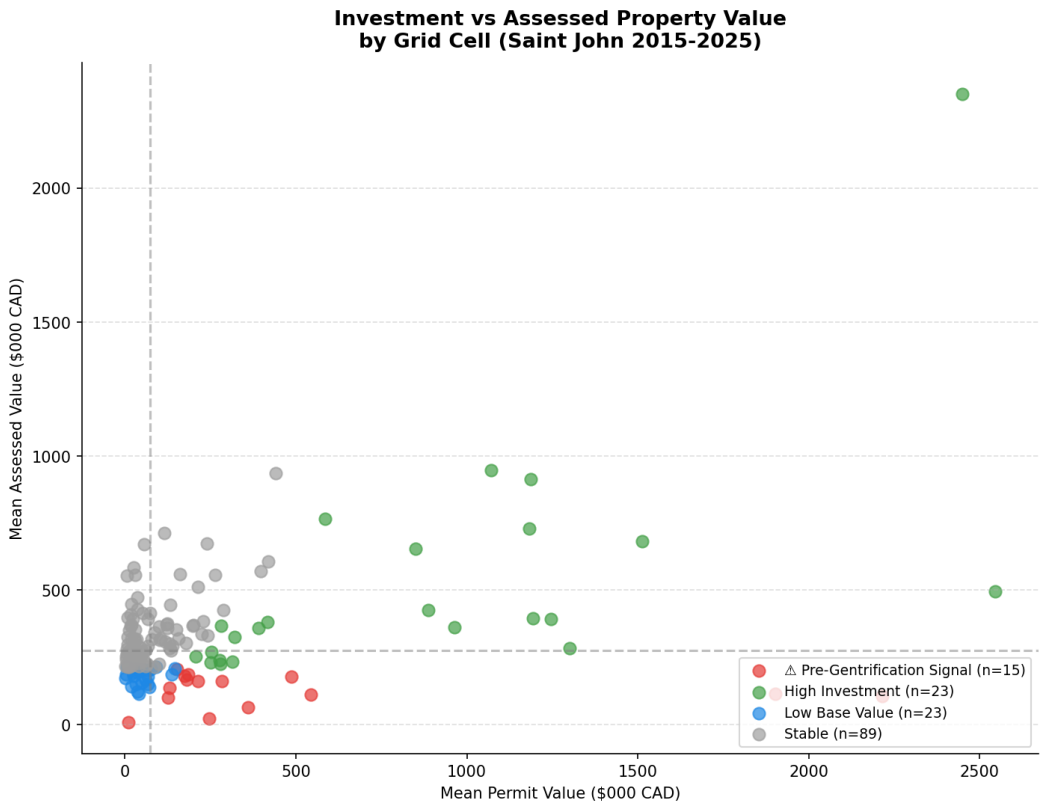

Figure 7: Investment versus assessment by grid cell. Cells in the upper-left region show high investment relative to still-low assessed values.

5 Results

5.1 Main Findings

The most important findings of the project are summarized in Table 2.

Table 2: Summary of Main Findings

Finding	**Result**
Raw permit records	30,287
Final cleaned permit records	1,098
Total investment represented	$255,986,293
Grid cells created	525
Active grid cells	151
K-Means result	6 clusters
DBSCAN result	1 cluster, 1 noise point
Significant Getis-Ord hotspots	1
Emerging temporal cells	1
Highest CAGR	71% (GRID_10_10)
Pre-gentrification cells (IAR)	15
Highest IAR	21.12 (GRID_03_16)

5.2 Top IAR Cells

The five strongest pre-gentrification signal cells were:

Table 3: Top 5 Pre-Gentrification Signal Cells

Cell	**IAR**	**Mean Permit $**	**Mean Assessment $**
GRID_03_16	21.12	2,215,000	104,893
GRID_15_24	16.92	1,903,000	112,477
GRID_17_22	10.85	247,000	22,767
GRID_04_18	5.77	360,000	62,419
GRID_09_07	4.88	544,050	111,456

The strongest example, GRID_03_16, had an average permit value over $2.2 million while the average assessed property value remained just above $100,000.

5.3 Additional Validation of Method Agreement

I also compared how the major methods aligned. The overlap was limited, which at first seemed surprising.

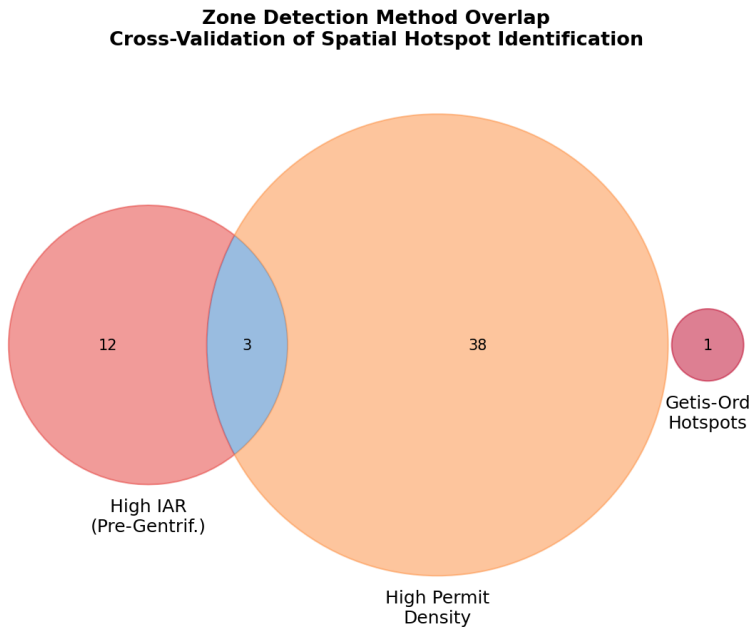

Figure 8: Comparison of different detection methods. The limited overlap suggests that the methods capture different aspects of neighborhood change.

This turned out to be meaningful:

- Getis-Ord detected an area where high values had already clustered,
- while IAR highlighted areas where investment was arriving before values had fully adjusted.

So the methods were not redundant; they were capturing different phases of change.

6 Discussion

6.1 What These Results Mean

The results suggest that building permits really can function as an early warning signal. In particular, the IAR metric appears to identify areas where investment is arriving ahead of official value recognition.

Saint John also appears to follow a more scattered pattern of urban change than the large cities often discussed in the literature. Rather than one large wave of gentrification spreading out from the core, the city seems to show pockets or islands of concentrated change.

6.2 Policy Relevance

The 15 identified pre-gentrification cells could be useful for planning in several ways:

- infrastructure capacity studies,

- early housing affordability responses,
- monitoring of waterfront redevelopment zones,
- and periodic urban change reporting.

GRID_10_10 is particularly important because it appears repeatedly in the analysis:

- highest permit count,
- very high average permit value,
- and the strongest temporal growth pattern.

6.3 Limitations

This project also has important limitations.

6.3.1 Data Reduction

The reduction from 30,287 permit records to 1,098 final records was substantial. This made the dataset cleaner, but it also creates the possibility that some legitimate unusual permits may have been removed along with duplicates and noisy records.

6.3.2 Grid Resolution

A 1 km grid was a practical choice, but smaller cells might reveal more detailed spatial patterns. On the other hand, smaller cells would also increase sparsity.

6.3.3 Assessment Lag

Property assessments do not instantly reflect market change. This means that IAR likely captures only part of the full gap between investment and market value.

6.3.4 Causation

This project identifies patterns and signals, not direct causal proof of gentrification or displacement. To study that more fully, I would need additional data such as rent changes, demographic shifts, or eviction records.

7 What I Learned

This project taught me several useful lessons.

First, data cleaning matters much more than I expected. The biggest challenge in the project was not running the algorithms but preparing the data in a trustworthy way.

Second, negative or limited results can still be useful. DBSCAN finding one large cluster was not a failure; it told me something about the spatial structure of Saint John.

Third, combining methods gave a much stronger interpretation than using one method alone. K-Means, DBSCAN, Getis-Ord, and IAR each highlighted different features of the same urban system.

Finally, visualization mattered a lot. The scatter plot between investment and assessment and the temporal heatmap were especially helpful for turning raw numerical patterns into something interpretable.

8 Future Work

There are several obvious directions for extending this project.

8.1 Predictive Modeling

A natural next step would be to build a predictive classifier that estimates which currently stable cells are most likely to become emerging cells in the next few years.

8.2 Additional Datasets

Adding rent data, demographic data, or sales data would strengthen the interpretation and help separate investment from actual displacement effects.

8.3 Comparison with Other Cities

It would also be useful to compare Saint John with Fredericton, Moncton, or Halifax to see whether the same patterns appear in other Maritime cities.

9 Conclusion

This project set out to test whether building permits can provide an earlier warning sign of neighborhood change than property assessments. Based on the results, I believe they can.

I identified:

- 15 pre-gentrification signal cells through the Investment-Assessment Ratio,
- 1 statistically significant hotspot,
- and 1 strongly emerging cell with 71% annual growth.

The main contribution of the project is the IAR metric, which gives a simple way to compare current investment with official property values. In Saint John, this metric helped identify areas where change appears to be happening before assessment values have fully responded.

More broadly, the project shows that permit data can be used for proactive urban analysis, not just retrospective reporting. If cities can detect neighborhood change earlier, they may be better able to respond with planning, infrastructure, and housing policies before the strongest pressures appear.

References

[1] Smith, N. (1979). Toward a theory of gentrification: A back to the city movement by capital, not people. *Journal of the American Planning Association*, 45(4), 538–548.

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[4] Behnisch, M., & Ultsch, A. (2009). Urban data mining: Spatiotemporal exploration of multidimensional data. *Building Research & Information*, 37(5–6), 520–532.

[5] City of Saint John. (2026). *Building Permits Open Data Collection*. Retrieved from https://catalogue-saintjohn.opendata.arcgis.com/

A Code and Data Availability

All code, processed data, figures, and documentation are available at:

https://github.com/MalekKaroui/renovation-roadmap

Repository contents include:

- `src/`: source code for the analysis pipeline and dashboard,
- `data/`: processed data outputs,
- `outputs/figures/`: all generated visualizations,
- `README.md`: setup and reproduction instructions.

B Software Environment

The project was implemented in Python 3.11 using the following libraries:

- pandas
- geopandas
- scikit-learn
- matplotlib
- libpysal
- esda
- streamlit

C Academic Integrity Statement

I used ChatGPT as a support tool for:

- debugging some GeoPandas CRS and path-related issues,
- improving LaTeX formatting and report structure,
- improving the formatting of terminal print statements and `echo` output used in the project runner and live demo scripts,
- and refining some presentation and documentation wording.

All substantive parts of the project remain my own work, including:

- data collection,
- preprocessing decisions,
- feature engineering,
- implementation and execution of the analysis pipeline,
- interpretation of results,
- visualization choices,
- and the final conclusions of the project.